🏗 Project Guidelines

INFORMATION REGARDING THE CLASS PROJECT

Project teams who follow the instructions in this document generally do well on their projects. It is best to start following the direction during the milestone stage. A secondary goal of the project is to provide practice in following instructions, preparing concise technical presentations, and writing a clear technical report.

As an integral part of this course, students are asked to find and analyze their own data for a project. This class project is designed as a team assignment because real-world data work rarely happens in isolation. Collaborating in a group helps students practice essential teamwork skills such as communication, task coordination, and integrating different perspectives into a unified analysis. Working with peers allows students to divide responsibilities, making it possible to explore a larger, more complex dataset than any individual could reasonably handle alone. This deepens your ability to interpret and draw meaningful insights from real data and mirrors the collaborative nature of professional data analysis, where success depends on combining technical skills with the ability to work effectively with others.

For each project, teams will complete two short milestone video presentations and one final written report. The milestones are intended to keep teams on schedule and provide feedback before the final project is completed. The final report should explain where and how the data were obtained, how the analysis proceeded, including any problems that were encountered and how they were solved, results of the analysis, and conclusions derived. Note that conciseness is one of the evaluation criteria. Suggestions and limitations for modeling should also be included. Figures, graphs, and tables should be included only when they are relevant and discussed in the report. The final report should not include raw computer output.

1. Choosing Your Data Sets

There are many sources of data on the Internet. Government agencies are some of the best sources. But be resourceful. It is better to find a data set from a process that interests you. Please do not use data that is available in a statistical software package (examples include: SAS, R, JMP, Minitab etc.), textbooks, or online repositories of heavily analyzed data sets. There are some interesting data sets on Kaggle.com. If you would like to use one of them or data from other similar repositories, you must go to the original source of the data and get up-to-date data. You may also use a non-confidential data set from your work or one of your research projects. Regardless of your source, you must provide the reference to the original data set within Milestone 1, Milestone 2, and the final report.

Other specifications for acceptable data sets:

- Your dataset should have at least 300 observations and at least 4 different explanatory variables.
- At least one of the explanatory variables should be categorical.

2. Project Milestones and Submission

The purpose of the milestones is to provide feedback to project teams and to keep them on track and on schedule. You should follow the guidelines given in this document in preparing the milestones. That way, the second milestone can be easily extended into the final project report.

Milestones should be submitted in **Canvas**. Please write the group name and list names of all team members in alphabetical order on the first slide. Only one member of the group should submit the milestone materials. I will set up Canvas so that comments and grades are applied to all group members.

Both milestone presentations must be recorded using Canvas Studio. The Canvas submission should include the presentation slides and the Canvas Studio video link. Each video presentation should be no longer than 5 minutes. All group members must present in each milestone video.

Make sure your final report is self-contained. It should include the important material from the milestones, corrected based on feedback when appropriate, and should not require the reader to refer back to previous milestones.

3. Project Grading

Part of your grade will depend on how well you follow the further instructions and guidelines in this document. Specifically, the projects will be evaluated based on the following equally-weighted criteria.

- Was Milestone 1 turned in on time and done according to instructions?
- Was Milestone 2 turned in on time and done according to instructions?
- Were the milestone videos clear, concise, and completed by all group members?
- Was the problem described adequately in terms of the goals of the analysis, the source of the data, the background of the process that generated the data, and so forth?
- Are the research questions clearly stated and appropriate for the data?
- Are there adequate graphics and figures, and have they been properly labeled with captions and described in the text?
- Were the tentative model identification and transformations done properly?
- Was the model fitting done well? Was there adequate discussion of sensible alternative models and an informative model-comparison table?
- Was the diagnostic checking done properly and completely? Were plots of residuals and Q-Q plots of the residuals properly interpreted?
- When prediction is included: were prediction intervals, forecast intervals, or hold-out results discussed appropriately?
- Was the final report well organized, clear, and concise?

- Were the conclusions of the analysis stated clearly?
- Were the R commands submitted on time and in a reproducible form?
- Was the project submitted on time?

4. Brief description of expectations for Milestones

Milestone #1 will be submitted as a short video presentation. The presentation should use no more than 3 slides and should include:

The group name and names of all team members, listed in alphabetical order.

- A clear description of the data source, including a reference or URL when the data are publicly available.
- A brief description of the dataset and the process that generated the data.
- Three research questions related to the data.
- One plot that is directly related to at least one of the research questions.

The Milestone #1 Canvas submission should include the slides and the Canvas Studio video link. All group members must present. The video should be no longer than 5 minutes.

Milestone #2 will also be submitted as a short video presentation. By this milestone, your group should have prepared and refined the three research questions, chosen a statistical analysis that is appropriate for the questions and data, and presented initial results. The presentation should include:

- The group name and names of all team members, listed in alphabetical order.
- The three research questions, revised if needed based on feedback from Milestone #1.
- A brief explanation of the statistical analysis selected for the project and why it is appropriate.
- Initial outcomes from the analysis, supported by relevant plots, tables, or short numerical summaries.
- A brief description of next steps needed before the final report.

The Milestone #2 Canvas submission should include the slides and the Canvas Studio video link. All group members must present. The video should be no longer than 5 minutes.

You will receive feedback about your data and your application of methodology for each of the milestones. There is a time gap between Milestone 1 and Milestone 2 so that initial proposals can be revised or inappropriate data can be replaced with something more suitable. I will include the rubric I use for project evaluation within each of the milestones.

5. Final Project Report and Submission

The final project should be submitted as a PDF report. The final report should be no longer than 5 pages, not including references or optional appendix material if specifically requested in Canvas. The report should be concise but complete.

- The final report should include:

- The project title, group name, and names of all team members.
- The data set name and data source, including a URL or reference when applicable.
- The three research questions.
- A brief description of the background and process that generated the data.
- Exploratory analysis that supports the goals of the project.
- The statistical analysis used to answer the research questions.
- Model fitting results, diagnostic checks, and discussion of limitations.
- Clear conclusions connected directly to the research questions.
- For the final project submission, submit the following files through Canvas as prompted:
- The final report in PDF format, no longer than 5 pages.
- Your R code file. This file should contain the R commands used in your project, in a logical order, so that the results can be reproduced if needed.
- A copy of your data file (.csv, .xls(x), or .txt), unless the data are fully public and the Canvas instructions state that a link is sufficient.

The R code file should correspond to the analyses in your final report. This commands file should not contain raw computer output or extraneous clutter. Do not submit an unedited R history file.

6. More detailed Instructions, Guidelines, and Suggestions for the Projects

- Briefly describe the process from which your realization came and give the source of your data. If your data are publicly available, provide a URL. That is, provide a specific reference so that someone else could find your data source.
- Explain the purpose of your analysis and modeling.
- Use appropriate scaling of your data, e.g., use 808.35 million dollars instead of 808,350,000 dollars.
- If you have any missing observations, explain the cause and explain how you dealt with the problem. For example, it may be appropriate to interpolate to fill in a missing value or remove missing observations.
- If fitting a regression model is part of your analysis, plot your response against all the proposed explanatory variables. Describe important features observed in these plots.
- If model selection for regression is part of your analysis, use automated model selection tools to tentatively identify one or more possible models for your data. Use R to estimate at least two different tentative models for your data. Try as many models as are needed until you find one that adequately describes your data, but include the results for no more than 5 different models in the final report. Do not present the result for models that are clearly inappropriate.
- Briefly describe the steps that you followed in arriving at your final model or models.
- Briefly describe the behavior of the process that you have been studying and explain what you have learned about this process by studying the data. This part should be no longer than necessary to convey the important information. Again, conciseness is one of the evaluation criteria.

- Once you have chosen whether to transform or not, stay with that choice for the complete analysis. Otherwise, modeling becomes too complicated. Include only the relevant plots in your final report, including original data plots and after-transformation plots if a transformation was used or seriously considered.
- Give your data files meaningful names, e.g., MilkPrice, instead of short abbreviations, e.g., m.
- Milestone slides and the final report should not include raw computer output. Include only relevant figures and computer output. Output is relevant only if you refer to it in your presentation or report.
- Divide the final report into logical sections and properly title and number each section. Also include page numbers.
- Your final report should be concise but complete. Your project will be graded on accuracy, completeness, the relevance of your graphical displays, tabular comparisons, and your statements, and on the report's organization, conciseness, and neatness.
- Whenever possible, use graphs plus a few words to make your points. Number your figures and use informative captions. Refer to the figures by name and number, e.g., "Figure 1 is a scatterplot of the response (Milk price) versus producer."
- Figures and tables should be integrated into the text and not placed at the end of the report. You can control the size of the figures, but do not make them so small that the axis labels are difficult to read. In tables, avoid scientific notation like 1.631e-03 and use 0.00163 instead. For p-values < 0.001, use < 0.001 because the exact magnitude of smaller numbers has little meaning.
- Use clear notation when formulas are needed, including Greek letters, subscripts, and mathematical symbols when appropriate.